

YASH TAKTE

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Education

Indiana University Bloomington

Master of Science in Data Science - GPA: 3.8/4.0

August 2024 - May 2026

Bloomington, Indiana, USA

Amrutvahini College of Engineering

Bachelor of Engineering in Computer Engineering - CGPA: 9.31/10.00

July 2019 - July 2023

Sangamner, Maharashtra, India

Technical Skills

Languages: Python, SQL, C++, JavaScript, HTML, CSS

Frameworks: TensorFlow, Keras, React, Express.js, Flask, Streamlit

Libraries: NumPy, Pandas, Matplotlib, Scikit-learn, Seaborn, OpenCV, Chart.js

Databases & Backend: MongoDB, Node.js

Tools & Platforms: Git, GitHub, Power BI, Tableau, Gephi, VS Code, Google Colab, Jupyter Notebook

Research Experience

Reddit-Based Analysis of Financial Anxiety During Economic Uncertainty | *Python, Scikit-learn*

April 2025

- Collected and cleaned 1,000+ Reddit comments from the post-pandemic economic period using PRAW and a custom NLP pipeline with lemmatization, tokenization, and stopword removal.
- Performed coherence-optimized LDA topic modeling to uncover seven key themes of financial anxiety, with interactive visualizations using pyLDAvis and supporting bar charts.
- Labeled sentiment using TextBlob polarity scores and trained a logistic regression sentiment classifier on TF-IDF features, achieving 90% accuracy and 0.95 AUC, with top predictive words visualized for Coping vs. Anxious classes.

Projects

AI-Powered Academic Recommendation Assistant | *Python, Streamlit, Power BI*

April 2025

- Built a chatbot-style academic assistant leveraging DeepSeek-V3 via Hugging Face API to generate context-aware, personalized course recommendations using prompt-engineered logic.
- Parsed structured catalog data and utilized DeepSeek-V3's built-in NLP capabilities to simulate academic advising, aligning recommendations with student goals, skills, and prerequisites.
- Designed a responsive Streamlit interface with custom CSS and embedded Power BI dashboards to visualize ratings, trends, and insights for transparent academic decision-making.

Spatiotemporal and Network Analysis of Global Science Exhibits | *Python, Pandas, Power BI*

April 2025

- Created a multi-layered visualization for SciMaps.org using Folium and Kepler.gl to explore 480+ global science exhibits, highlighting venue growth, organizer networks, and geographic expansion.
- Designed a preprocessing pipeline to standardize location names, calculate event duration, apply jittering to reduce map overlap, and integrate organizer-edge relationships for spatiotemporal filtering.
- Revealed collaboration hubs and outreach trends through animated timelines, network overlays, and a Power BI dashboard tailored for client-driven insight delivery.

Apple Leaf Disease Classification with Deep Learning | *Python, Scikit-learn, TensorFlow*

November 2024

- Processed 1,870 images across three categories including Apple Scab, Black Rot, and Healthy leaves using label encoding and one-hot encoding for multi-class classification.
- Applied PCA preserving 90% variance and visualized data using t-SNE, LLE, and MDS for dimensionality reduction and pattern analysis.
- Trained a CNN-based classification model with early stopping achieving 98.6% training accuracy and 87.2% validation accuracy.

Happiness Data Predictive Modeling | *Python, Pandas, Matplotlib, Scikit-learn, Seaborn*

September 2024

- Developed and tuned nine regression models to predict happiness scores, achieving a test MSE of 0.2219 using polynomial regression with Lasso, and evaluated performance using K-Fold Cross-Validation and hyperparameter tuning.
- Cleaned socioeconomic data by imputing missing values, removing 15% outliers using the IQR method, and standardizing features to improve model performance and data consistency.
- Created 10+ visualizations and performed correlation analysis to identify key drivers of happiness and uncover trends like positive links with income and health, and negative impacts from corruption and negative emotions.